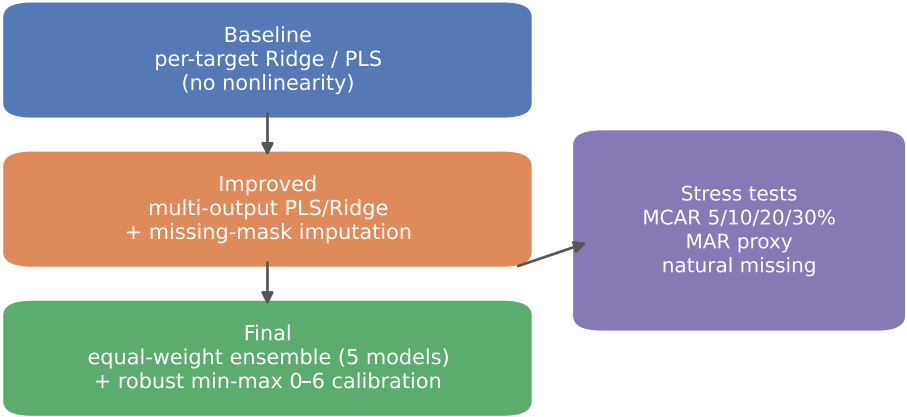Q1 modelling framework